

Package ‘countryscales’

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Title Country Scales

Version 0.3.0

Description Format numbers, percentages and currencies, and label 'ggplot2' axes, using country- or locale-specific conventions such as the thousands separator, decimal mark, currency symbol placement, and sign placement. Locale data is sourced from the Unicode Common Locale Data Repository (CLDR, <<https://cldr.unicode.org>>) via the 'i18n' package, covering several hundred locales in addition to dedicated helpers for Germany, Switzerland and the United States.

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Encoding UTF-8

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Imports ggplot2 (>= 3.5.0), plyr, rlang, scales, i18n

Depends R (>= 3.5.0)

LazyData true

URL <https://github.com/treconom/countryscales>,
<https://treconom.github.io/countryscales/>

BugReports <https://github.com/treconom/countryscales/issues>

Suggests covr, knitr, ragg, rmarkdown, testthat (>= 3.0.0)

VignetteBuilder knitr

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Config/Needs/website ragg

NeedsCompilation no

Author Stefan Moog [aut, cre],
scales authors [aut] (Authors of included code from the scales package),
Posit, PBC [cph] (Copyright holder of included code from the scales package)

Maintainer Stefan Moog <moogs@gmx.de>

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countries	<i>Countries</i>
-----------	------------------

Description

A data.frame of country names, language codes, country codes and locales

Usage

countries

Format

An object of class tbl_df (inherits from tbl, data.frame) with 377 rows and 4 columns.

Source

<https://cdn.simplelocalize.io/public/v1/locales>

currencies	<i>A table with localized currency attributes and descriptors</i>
------------	---

Description

Re-Imported from i18n to silent R CMD check errors.

Usage

```
currencies
```

Format

An object of class `tbl_df` (inherits from `tbl`, `data.frame`) with 175070 rows and 7 columns.

Source

<https://rich-iannone.github.io/i18n/>

demo_number	<i>ggplot2 demo</i>
-------------	---------------------

Description

ggplot2 demo

Usage

```
demo_number(x, scale_name, ...)
```

Arguments

<code>x</code>	a numeric
<code>scale_name</code>	scale name
<code>...</code>	further arguments to be passed to the scale

Value

A ggplot object, invisibly printed as a demo of the requested scale.

Examples

```
demo_number(c(32, 212), scale_name = "number_locale")
```

g20	<i>G20 countries</i>
-----	----------------------

Description

A data.frame of names, codes and locales for G20 countries

Usage

```
g20
```

Format

An object of class tbl_df (inherits from tbl, data.frame) with 19 rows and 5 columns.

Source

<https://en.wikipedia.org/wiki/G20>, <https://cdn.simplelocalize.io/public/v1/locales>

gapminder	<i>Gapminder Dataset</i>
-----------	--------------------------

Description

A dataset containing data on GDP per capita, life expectancy and population size of 183 countries for 1800 to 2015

Usage

```
gapminder
```

```
gapminder15
```

Format

A data frame with 14,823 rows and 7 variables:

country country name

code country code

region world region

year year

life_exp life expectancy at birth, in years

gdp_per_cap GDP per capita, in dollars PPP

pop population size, in persons

An object of class tbl_df (inherits from tbl, data.frame) with 183 rows and 7 columns.

Source

<https://www.gapminder.org/>

label-ar

Argentine Style Formatting of Numbers

Description

The label_xxx_ar family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_ar(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
label_percent_ar(  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
label_currency_ar(  
  accuracy = 1,  
  scale = 1,  
  currency = "ARS",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_ar formats numbers in decimal format.
- label_percent_ar formats numbers as percentages.
- label_currency_ar formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_ar()(c(-1e6, 0, 1e6))
label_percent_ar(accuracy = .01)(c(-1, 0, 1))
label_currency_ar(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-at*Austrian Style Formatting of Numbers*

Description

The label_xxx_at family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_at(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_percent_at(  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_currency_at(  
  accuracy = 1,  
  scale = 1,  
  currency = "EUR",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum
----------	--

	number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: <code>x</code> will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>base::format()</code> .
currency	currency symbol

Details

- `label_number_at` formats numbers in decimal format.
- `label_percent_at` formats numbers as percentages.
- `label_currency_at` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_at()(c(-1e6, 0, 1e6))
label_percent_at(accuracy = .01)(c(-1, 0, 1))
label_currency_at(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-au

Australian Style Formatting of Numbers

Description

The `label_xxx_au` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```

label_number_au(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_percent_au(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_currency_au(
  accuracy = 1,
  scale = 1,
  currency = "AUD",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.

big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_br formats numbers in decimal format.
- label_percent_br formats numbers as percentages.
- label_currency_br formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_br()(c(-1e6, 0, 1e6))
label_percent_br(accuracy = .01)(c(-1, 0, 1))
label_currency_br(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-br

Brazilian Style Formatting of Numbers

Description

The label_xxx_br family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_br(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```

)

label_percent_br(
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

label_currency_br(
    accuracy = 1,
    scale = 1,
    currency = "BRL",
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- `label_number_br` formats numbers in decimal format.
- `label_percent_br` formats numbers as percentages.
- `label_currency_br` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_br()(c(-1e6, 0, 1e6))
label_percent_br(accuracy = .01)(c(-1, 0, 1))
label_currency_br(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-ca

Canadian Style Formatting of Numbers

Description

The `label_xxx_ca` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_ca(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_ca(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
```

```

    ...
  )

  label_currency_ca(
    accuracy = 1,
    scale = 1,
    currency = "CAD",
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
  )

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_ca formats numbers in decimal format.
- label_percent_ca formats numbers as percentages.
- label_currency_ca formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_ca()(c(-1e6, 0, 1e6))
label_percent_ca(accuracy = .01)(c(-1, 0, 1))
label_currency_ca(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-ch*Swiss Style Formatting of Numbers*

Description

The label_xxx_xxx family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_ch(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_ch(
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_currency_ch(
  accuracy = 1,
  scale = 1,
  currency = "CHF",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
```

```

    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_ch formats numbers in decimal format.
- label_percent_ch formats numbers as percentages.
- label_currency_ch formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```

require(scales)

demo_continuous(c(-1e6, 1e6),
  labels = label_number_ch()
)
demo_continuous(c(-1, 1),
  labels = label_percent_ch(accuracy = .01)
)
demo_continuous(c(-1, 1),
  labels = label_currency_ch(accuracy = .1)
)

```

label-cn*Chinese Style Formatting of Numbers*

Description

The `label_xxx_cn` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_cn(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_percent_cn(  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_currency_cn(  
  accuracy = 1,  
  scale = 1,  
  currency = "CNY",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum
-----------------------	---

	number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>base::format()</code> .
currency	currency symbol

Details

- `label_number_cn` formats numbers in decimal format.
- `label_percent_cn` formats numbers as percentages.
- `label_currency_cn` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_cn(c(-1e6, 0, 1e6))
label_percent_cn(accuracy = .01)(c(-1, 0, 1))
label_currency_cn(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-de

German Style Formatting of Numbers

Description

The `label_xxx_xxx` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_de(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
label_percent_de(  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
label_currency_de(  
  accuracy = 1,  
  scale = 1,  
  currency = "EUR",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.

big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- `label_number_de` formats numbers in decimal format.
- `label_percent_de` formats numbers as percentages.
- `label_currency_de` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_de()(c(-1e6, 0, 1e6))
label_percent_de(accuracy = .01)(c(-1, 0, 1))
label_currency_de(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-dk

Danish Style Formatting of Numbers

Description

The `label_xxx_dk` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_dk(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```

label_percent_dk(
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

label_currency_dk(
    accuracy = 1,
    scale = 1,
    currency = "DKK",
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_dk formats numbers in decimal format.

- `label_percent_dk` formats numbers as percentages.
- `label_currency_dk` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_dk()(c(-1e6, 0, 1e6))
label_percent_dk(accuracy = .01)(c(-1, 0, 1))
label_currency_dk(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-es

Spanish Style Formatting of Numbers

Description

The `label_xxx_es` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_es(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_percent_es(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```

label_currency_es(
  accuracy = 1,
  scale = 1,
  currency = "EUR",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_es formats numbers in decimal format.
- label_percent_es formats numbers as percentages.
- label_currency_es formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_es()(c(-1e6, 0, 1e6))
label_percent_es(accuracy = .01)(c(-1, 0, 1))
label_currency_es(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-fi

Finnish Style Formatting of Numbers

Description

The label_xxx-fi family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number-fi(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent-fi(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_currency-fi(
  accuracy = 1,
  scale = 1,
  currency = "EUR",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
```

```

    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number-fi formats numbers in decimal format.
- label_percent-fi formats numbers as percentages.
- label_currency-fi formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```

# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number-fi()(c(-1e6, 0, 1e6))
label_percent-fi(accuracy = .01)(c(-1, 0, 1))
label_currency-fi(accuracy = .1)(c(-1e6, 0, 1e6))

```

label-fr*French Style Formatting of Numbers*

Description

The `label_xxx_fr` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_fr(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_percent_fr(  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_currency_fr(  
  accuracy = 1,  
  scale = 1,  
  currency = "EUR",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum
-----------------------	---

	number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>base::format()</code> .
currency	currency symbol

Details

- `label_number_fr` formats numbers in decimal format.
- `label_percent_fr` formats numbers as percentages.
- `label_currency_fr` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_fr()(c(-1e6, 0, 1e6))
label_percent_fr(accuracy = .01)(c(-1, 0, 1))
label_currency_fr(accuracy = .1)(c(-1e6, 0, 1e6))
```

Description

The `label_xxx_gb` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```

label_number_gb(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_percent_gb(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_currency_gb(
  accuracy = 1,
  scale = 1,
  currency = "GBP",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.

big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- `label_number_gb` formats numbers in decimal format.
- `label_percent_gb` formats numbers as percentages.
- `label_currency_gb` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_gb()(c(-1e6, 0, 1e6))
label_percent_gb(accuracy = .01)(c(-1, 0, 1))
label_currency_gb(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-id

Indonesian Style Formatting of Numbers

Description

The `label_xxx_id` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_id(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...)
```

```

)

label_percent_id(
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

label_currency_id(
    accuracy = 1,
    scale = 1,
    currency = "IDR",
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- `label_number_id` formats numbers in decimal format.
- `label_percent_id` formats numbers as percentages.
- `label_currency_id` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_id()(c(-1e6, 0, 1e6))
label_percent_id(accuracy = .01)(c(-1, 0, 1))
label_currency_id(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-ie

Irish Style Formatting of Numbers

Description

The `label_xxx_ie` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_ie(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_ie(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
```

```

    ...
  )

  label_currency_ie(
    accuracy = 1,
    scale = 1,
    currency = "EUR",
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
  )

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_ie formats numbers in decimal format.
- label_percent_ie formats numbers as percentages.
- label_currency_ie formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_ie()(c(-1e6, 0, 1e6))
label_percent_ie(accuracy = .01)(c(-1, 0, 1))
label_currency_ie(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-it

Italian Style Formatting of Numbers

Description

The `label_xxx_it` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_it(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_it(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_currency_it(
  accuracy = 1,
  scale = 1,
  currency = "EUR",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
```

```

    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_it formats numbers in decimal format.
- label_percent_it formats numbers as percentages.
- label_currency_it formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```

# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_it()(c(-1e6, 0, 1e6))
label_percent_it(accuracy = .01)(c(-1, 0, 1))
label_currency_it(accuracy = .1)(c(-1e6, 0, 1e6))

```

label-jp*Japanese Style Formatting of Numbers*

Description

The `label_xxx_jp` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_jp(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_percent_jp(  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_currency_jp(  
  accuracy = 1,  
  scale = 1,  
  currency = "JPY",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum
-----------------------	---

	number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_jp formats numbers in decimal format.
- label_percent_jp formats numbers as percentages.
- label_currency_jp formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_jp()(c(-1e6, 0, 1e6))
label_percent_jp(accuracy = .01)(c(-1, 0, 1))
label_currency_jp(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-kr	South Korean Style Formatting of Numbers
----------	--

Description

The label_xxx_kr family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```

label_number_kr(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_percent_kr(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_currency_kr(
  accuracy = 1,
  scale = 1,
  currency = "KRW",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.

big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_kr formats numbers in decimal format.
- label_percent_kr formats numbers as percentages.
- label_currency_kr formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_kr()(c(-1e6, 0, 1e6))
label_percent_kr(accuracy = .01)(c(-1, 0, 1))
label_currency_kr(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-locale

Formatting numbers

Description

The label_xxx_locale family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_locale(
  accuracy = NULL,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  locale = "en-US",
  trim = TRUE,
```

```

    ...
)

label_percent_locale(
  accuracy = NULL,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  locale = "en-US",
  trim = TRUE,
  ...
)

label_currency_locale(
  accuracy = NULL,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  p_sep_by = NULL,
  n_sep_by = NULL,
  currency = "USD",
  locale = "en-US",
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
locale	locale string. Defaults to "en-US"

trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>base::format()</code> .
p_sep_by	separator between currency symbol and positive monetary value
n_sep_by	separator between currency symbol and negative monetary value
currency	currency symbol

Details

- `label_number_locale` formats numbers in decimal format.
- `label_percent_locale` formats numbers as percentages.
- `label_currency_locale` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_locale(locale = "fr-FR")(c(-1e6, 0, 1e6))
label_percent_locale(locale = "it-IT", accuracy = .01)(c(-1, 0, 1))
label_currency_locale(
  locale = "ja-JP", accuracy = .1,
  currency = "JPY"
)(c(-1e4, 0, 1e4))
```

label-mx	<i>Mexican Style Formatting of Numbers</i>
----------	--

Description

The `label_xxx_mx` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_mx(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```

)

label_percent_mx(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_currency_mx(
  accuracy = 1,
  scale = 1,
  currency = "MXN",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- `label_number_mx` formats numbers in decimal format.
- `label_percent_mx` formats numbers as percentages.
- `label_currency_mx` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_mx()(c(-1e6, 0, 1e6))
label_percent_mx(accuracy = .01)(c(-1, 0, 1))
label_currency_mx(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-no

Norwegian Style Formatting of Numbers

Description

The `label_xxx_no` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_no(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_no(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
```

```

    ...
  )

  label_currency_no(
    accuracy = 1,
    scale = 1,
    currency = "NOK",
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
  )

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_no formats numbers in decimal format.
- label_percent_no formats numbers as percentages.
- label_currency_no formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_no()(c(-1e6, 0, 1e6))
label_percent_no(accuracy = .01)(c(-1, 0, 1))
label_currency_no(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-pt

Portuguese Style Formatting of Numbers

Description

The label_xxx_pt family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_pt(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_pt(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_currency_pt(
  accuracy = 1,
  scale = 1,
  currency = "EUR",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
```

```

    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_pt formats numbers in decimal format.
- label_percent_pt formats numbers as percentages.
- label_currency_pt formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```

# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_pt()(c(-1e6, 0, 1e6))
label_percent_pt(accuracy = .01)(c(-1, 0, 1))
label_currency_pt(accuracy = .1)(c(-1e6, 0, 1e6))

```

Description

The `label_xxx_ru` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_ru(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_percent_ru(  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_currency_ru(  
  accuracy = 1,  
  scale = 1,  
  currency = "RUB",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum
-----------------------	---

	number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_ru formats numbers in decimal format.
- label_percent_ru formats numbers as percentages.
- label_currency_ru formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_ru()(c(-1e6, 0, 1e6))
label_percent_ru(accuracy = .01)(c(-1, 0, 1))
label_currency_ru(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-sa	<i>Saudi Style Formatting of Numbers</i>
----------	--

Description

The label_xxx_sa family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```

label_number_sa(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_percent_sa(
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_currency_sa(
  accuracy = 1,
  scale = 1,
  currency = "SAR",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.

big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_sa formats numbers in decimal format.
- label_percent_sa formats numbers as percentages.
- label_currency_sa formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_sa()(c(-1e6, 0, 1e6))
label_percent_sa(accuracy = .01)(c(-1, 0, 1))
label_currency_sa(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-se

Swedish Style Formatting of Numbers

Description

The label_xxx_se family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_se(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```

)

label_percent_se(
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

label_currency_se(
  accuracy = 1,
  scale = 1,
  currency = "SEK",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_se formats numbers in decimal format.
- label_percent_se formats numbers as percentages.
- label_currency_se formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_se()(c(-1e6, 0, 1e6))
label_percent_se(accuracy = .01)(c(-1, 0, 1))
label_currency_se(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-tr

Turkish Style Formatting of Numbers

Description

The label_xxx_tr family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_tr(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_tr(
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
```

```

    ...
)

label_currency_tr(
  accuracy = 1,
  scale = 1,
  currency = "TRY",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_tr formats numbers in decimal format.
- label_percent_tr formats numbers as percentages.
- label_currency_tr formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_tr()(c(-1e6, 0, 1e6))
label_percent_tr(accuracy = .01)(c(-1, 0, 1))
label_currency_tr(accuracy = .1)(c(-1e6, 0, 1e6))
```

label-us	<i>US Style Formatting of Numbers</i>
----------	---------------------------------------

Description

The `label_xxx_xxx` family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_us(
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_percent_us(
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
label_currency_us(
  accuracy = 1,
  scale = 1,
  currency = "USD",
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
```

```

    decimal.mark = NULL,
    trim = TRUE,
    ...
)

```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to base::format() .
currency	currency symbol

Details

- label_number_us formats numbers in decimal format.
- label_percent_us formats numbers as percentages.
- label_currency_us formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```

require(scales)

demo_continuous(c(-1e6, 1e6),
  labels = label_number_us()
)
demo_continuous(c(-1, 1),
  labels = label_percent_us(accuracy = .1)
)
demo_continuous(c(-1, 1),
  labels = label_currency_us(accuracy = .1)
)

```

label-za

South African Style Formatting of Numbers

Description

The label_xxx_za family of functions makes it easy to format numbers in decimal format, as percentages or as currencies.

Usage

```
label_number_za(  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_percent_za(  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
label_currency_za(  
  accuracy = 1,  
  scale = 1,  
  currency = "ZAR",  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

Arguments

accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum
----------	--

	number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: <code>x</code> will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>base::format()</code> .
currency	currency symbol

Details

- `label_number_za` formats numbers in decimal format.
- `label_percent_za` formats numbers as percentages.
- `label_currency_za` formats numbers as currencies.

Value

A labelling function that takes a numeric vector and returns a character vector.

Examples

```
# rendered directly (rather than via scales::demo_continuous()) because
# several locales' group/currency signs aren't in the check device's font
label_number_za()(c(-1e6, 0, 1e6))
label_percent_za(accuracy = .01)(c(-1, 0, 1))
label_currency_za(accuracy = .1)(c(-1e6, 0, 1e6))
```

locales	<i>Locales</i>
---------	----------------

Description

A dataset containing specification of formatting styles to label numbers, percentages and currencies for 574 locales based on CLDR (Common Locale Data Repository) data provided by the [i18n](#) package.

Usage

locales

Format

An object of class `tbl_df` (inherits from `tbl`, `data.frame`) with 764 rows and 46 columns.

Source

<https://rich-iannone.github.io/i18n/>

scale-ar

Argentine Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_ar(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_ar(
  name = waiver(),
```

```
breaks = waiver(),
minor_breaks = waiver(),
guide = waiver(),
n.breaks = NULL,
labels = waiver(),
limits = NULL,
expand = waiver(),
oob = censor,
na.value = NA_real_,
trans = "identity",
position = "bottom",
sec.axis = waiver(),
accuracy = 1,
scale = 100,
prefix = "",
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
trim = TRUE,
...
)
```

```
scale_x_currency_ar(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "ARS",
  trim = TRUE,
  ...
)
```

```
scale_y_number_ar(
```

```
    name = waiver(),
    breaks = waiver(),
    minor_breaks = waiver(),
    guide = waiver(),
    n.breaks = NULL,
    labels = waiver(),
    limits = NULL,
    expand = waiver(),
    oob = censor,
    na.value = NA_real_,
    trans = "identity",
    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
  )
```

```
scale_y_percent_ar(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_y_currency_ar(
```

```

name = waiver(),
breaks = waiver(),
minor_breaks = waiver(),
guide = waiver(),
n.breaks = NULL,
labels = waiver(),
limits = NULL,
expand = waiver(),
oob = censor,
na.value = NA_real_,
trans = "identity",
position = "left",
sec.axis = waiver(),
accuracy = 1,
scale = 1,
prefix = NULL,
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
currency = "ARS",
trim = TRUE,
...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.

n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.

prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to <code>scale_(x y)_continuous()</code>
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_ar()
scale_x_percent_ar()
scale_x_currency_ar()
```

scale-at

Austrian Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_at(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_percent_at(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_at(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_at(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)

scale_y_percent_at(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_at(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_at()
scale_x_percent_at()
scale_x_currency_at()
```

scale-au

Australian Style Positional Scales

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_au(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_au(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_au(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "AUD",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_au(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_au(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_au(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "AUD",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_au()
scale_x_percent_au()
scale_x_currency_au()
```

scale-br

Brazilian Style Positional Scales

Description

The scale_XXX family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_br(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_x_percent_br(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_br(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "BRL",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_br(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)  
  
scale_y_percent_br(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_y_currency_br(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "BRL",  
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_br()
scale_x_percent_br()
scale_x_currency_br()
```

scale-ca

Canadian Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_ca(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_ca(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_ca(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "CAD",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_ca(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_ca(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_ca(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "CAD",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_ca()
scale_x_percent_ca()
scale_x_currency_ca()
```

scale-ch

Swiss Style Positional Scales

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_ch(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_y_number_ch(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_percent_ch(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_ch(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_ch(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "CHF",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_currency_ch(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "CHF",  
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
demo_number(c(-1e6, 1e6), scale_name = "number_ch")
scale_x_number_ch()
demo_number(c(-1, 1), scale_name = "percent_ch")
scale_x_percent_ch()
demo_number(c(-1e4, 1e4), scale_name = "currency_ch")
scale_x_currency_ch()
```

scale-cn

Chinese Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_cn(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_x_percent_cn(
```

```
    name = waiver(),
    breaks = waiver(),
    minor_breaks = waiver(),
    guide = waiver(),
    n.breaks = NULL,
    labels = waiver(),
    limits = NULL,
    expand = waiver(),
    oob = censor,
    na.value = NA_real_,
    trans = "identity",
    position = "bottom",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
  )
```

```
scale_x_currency_cn(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "CNY",
  trim = TRUE,
  ...
)
```

```
scale_y_number_cn(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_cn(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_cn(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "CNY",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with <code>NA</code>. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of <code>transform</code> .
position	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_cn()
scale_x_percent_cn()
scale_x_currency_cn()
```

scale-de

German Style Positional Scales

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_de(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_y_number_de(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_percent_de(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_de(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_de(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_currency_de(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
demo_number(c(-1e6, 1e6), scale_name = "number_de")
scale_x_number_de()
demo_number(c(-1, 1), scale_name = "percent_de")
scale_x_percent_de()
# not rendered via demo_number(): the euro sign isn't in the check
# device's font, so only the (unrendered) scale object is shown
scale_x_currency_de()
```

scale-dk

Danish Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_dk(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_dk(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_dk(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "DKK",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_dk(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_dk(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_dk(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "DKK",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_dk()
scale_x_percent_dk()
scale_x_currency_dk()
```

scale-es	<i>Spanish Style Positional Scales</i>
----------	--

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_es(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_x_percent_es(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_es(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_es(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)

scale_y_percent_es(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_es(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_es()
scale_x_percent_es()
scale_x_currency_es()
```

scale-fi

Finnish Style Positional Scales

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_fi(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_fi(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_fi(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_fi(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_fi(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency-fi(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with <code>NA</code>. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of <code>transform</code> .
position	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_fi()
scale_x_percent_fi()
scale_x_currency_fi()
```

scale-fr

French Style Positional Scales

Description

The scale_XXX family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_fr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_x_percent_fr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_fr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_fr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)

scale_y_percent_fr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_fr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_fr()
scale_x_percent_fr()
scale_x_currency_fr()
```

scale-gb

British Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_gb(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_gb(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_gb(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "GBP",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_gb(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_gb(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_gb(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "GBP",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_gb()
scale_x_percent_gb()
scale_x_currency_gb()
```

scale-id	<i>Indonesian Style Positional Scales</i>
----------	---

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_id(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_x_percent_id(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_id(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "IDR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_id(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)

scale_y_percent_id(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_id(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "IDR",
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_id()
scale_x_percent_id()
scale_x_currency_id()
```

scale-ie

Irish Style Positional Scales

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_ie(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_ie(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_ie(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_ie(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_ie(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_ie(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_ie()
scale_x_percent_ie()
scale_x_currency_ie()
```

scale-it

Italian Style Positional Scales

Description

The scale_XXX_XXX family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_it(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_x_percent_it(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_it(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "EUR",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_it(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)

scale_y_percent_it(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_it(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_it()
scale_x_percent_it()
scale_x_currency_it()
```

scale-jp

Japanese Style Positional Scales

Description

The scale_xxx_xxx family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_jp(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_jp(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_jp(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "JPY",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_jp(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_y_percent_jp(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```

scale_y_currency_jp(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "JPY",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.

guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with <code>NA</code>. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of <code>transform</code> .
position	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.

scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_jp()
scale_x_percent_jp()
scale_x_currency_jp()
```

scale-kr

South Korean Style Positional Scales

Description

The scale_XXX family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_kr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)  
  
scale_x_percent_kr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = "",  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_currency_kr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  currency = "KRW",  
  trim = TRUE,  
  ...  
)
```

```
scale_y_number_kr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
)

scale_y_percent_kr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_kr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "KRW",
  trim = TRUE,
```

```
    ...  
  )
```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts <code>rlang</code> lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts <code>rlang</code> lambda function notation.
limits	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum

	• A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range. • scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_kr()
scale_x_percent_kr()
scale_x_currency_kr()
```

scale-locale

Formatted Positional Scales

Description

The `scale_x/y_XXX_XXX` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_locale(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  locale = "en-US",
  trim = TRUE,
  ...
)
```

```
scale_y_number_locale(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "left",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  locale = "en-US",  
  trim = TRUE,  
  ...  
)
```

```
scale_x_percent_locale(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 100,  
  prefix = NULL,  
  suffix = NULL,  
  big.mark = NULL,  
  decimal.mark = NULL,  
  locale = "en-US",  
  trim = TRUE,
```

```
    ...  
  )  
  
  scale_y_percent_locale(  
    name = waiver(),  
    breaks = waiver(),  
    minor_breaks = waiver(),  
    guide = waiver(),  
    n.breaks = NULL,  
    labels = waiver(),  
    limits = NULL,  
    expand = waiver(),  
    oob = censor,  
    na.value = NA_real_,  
    trans = "identity",  
    position = "left",  
    sec.axis = waiver(),  
    accuracy = 1,  
    scale = 100,  
    prefix = NULL,  
    suffix = NULL,  
    big.mark = NULL,  
    decimal.mark = NULL,  
    locale = "en-US",  
    trim = TRUE,  
    ...  
  )  
  
  scale_x_currency_locale(  
    name = waiver(),  
    breaks = waiver(),  
    minor_breaks = waiver(),  
    guide = waiver(),  
    n.breaks = NULL,  
    labels = waiver(),  
    limits = NULL,  
    expand = waiver(),  
    oob = censor,  
    na.value = NA_real_,  
    trans = "identity",  
    position = "bottom",  
    sec.axis = waiver(),  
    accuracy = 1,  
    scale = 1,  
    prefix = NULL,  
    suffix = NULL,  
    big.mark = NULL,  
    decimal.mark = NULL,
```

```

    locale = "en-US",
    currency = "USD",
    trim = TRUE,
    ...
)

scale_y_currency_locale(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  locale = "en-US",
  currency = "USD",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts <code>rlang</code> lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks

	• <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang <code>lambda</code> function notation. When the function has two arguments, it will be given the limits and major breaks.
<code>guide</code>	A function used to create a guide or its name. See <code>guides()</code> for more information.
<code>n.breaks</code>	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
<code>labels</code>	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang <code>lambda</code> function notation.
<code>limits</code>	One of: • <code>NULL</code> to use the default scale range • A numeric vector of length two providing limits of the scale. Use <code>NA</code> to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang <code>lambda</code> function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see <code>coord_cartesian()</code>).
<code>expand</code>	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function <code>expansion()</code> to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
<code>oob</code>	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang <code>lambda</code> function notation. • The default (<code>scales:::censor()</code>) replaces out of bounds values with <code>NA</code>. • <code>scales:::squish()</code> for squishing out of bounds values into range. • <code>scales:::squish_infinite()</code> for squishing infinite values into range.
<code>na.value</code>	Missing values will be replaced with this value.
<code>trans</code>	[Deprecated] Deprecated in favour of <code>transform</code> .
<code>position</code>	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.

<code>sec.axis</code>	<code>sec_axis()</code> is used to specify a secondary axis.
<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
<code>scale</code>	A scaling factor: <code>x</code> will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
<code>prefix</code>	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
<code>suffix</code>	Additional text to display after the number.
<code>big.mark</code>	Character used between every 3 digits to separate thousands. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>decimal.mark</code>	The character to be used to indicate the numeric decimal point. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>locale</code>	locale string. Defaults to <code>"en-US"</code>
<code>trim</code>	Logical, if <code>FALSE</code> , values are right-justified to a common width (see <code>base::format()</code>).
<code>...</code>	Other arguments passed on to <code>scale_(x y)_continuous()</code>
<code>currency</code>	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A `ggplot2` scale, for use in a `+`-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_locale(locale = "fr-FR")
scale_x_percent_locale(locale = "it-IT")
scale_x_currency_locale(locale = "ja-JP", currency = "JPY")
```

scale-mx

Mexican Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_mx(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_mx(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_currency_mx(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "MXN",
  trim = TRUE,
  ...
)

scale_y_number_mx(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```
    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_percent_mx(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_mx(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

position = "left",
sec.axis = waiver(),
accuracy = 1,
scale = 1,
prefix = NULL,
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
currency = "MXN",
trim = TRUE,
...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks)

	• An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang <code>lambda</code> function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang <code>lambda</code> function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see <code>coord_cartesian()</code>).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function <code>expansion()</code> to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang <code>lambda</code> function notation. • The default (<code>scales::censor()</code>) replaces out of bounds values with NA. • <code>scales::squish()</code> for squishing out of bounds values into range. • <code>scales::squish_infinite()</code> for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	<code>sec_axis()</code> is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .

trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>scale_(x y)_continuous()</code>
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_mx()
scale_x_percent_mx()
scale_x_currency_mx()
```

scale-no	<i>Norwegian Style Positional Scales</i>
----------	--

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_no(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_percent_no(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_x_currency_no(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    currency = "NOK",
    trim = TRUE,
    ...
)

scale_y_number_no(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_percent_no(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_currency_no(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "NOK",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for po-

	sition scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • NULL for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code>. Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range.

	• <code>scales::squish_infinite()</code> for squishing infinite values into range.
<code>na.value</code>	Missing values will be replaced with this value.
<code>trans</code>	[Deprecated] Deprecated in favour of <code>transform</code> .
<code>position</code>	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
<code>sec.axis</code>	<code>sec_axis()</code> is used to specify a secondary axis.
<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
<code>scale</code>	A scaling factor: <code>x</code> will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
<code>prefix</code>	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
<code>suffix</code>	Additional text to display after the number.
<code>big.mark</code>	Character used between every 3 digits to separate thousands. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>decimal.mark</code>	The character to be used to indicate the numeric decimal point. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>trim</code>	Logical, if <code>FALSE</code> , values are right-justified to a common width (see <code>base::format()</code>).
<code>...</code>	Other arguments passed on to <code>scale_(x y)_continuous()</code>
<code>currency</code>	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A `ggplot2` scale, for use in a `+`-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_no()
scale_x_percent_no()
scale_x_currency_no()
```

scale-pt*Portuguese Style Positional Scales*

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_pt(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_percent_pt(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_currency_pt(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "EUR",
  trim = TRUE,
  ...
)

scale_y_number_pt(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```
    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_percent_pt(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_pt(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

position = "left",
sec.axis = waiver(),
accuracy = 1,
scale = 1,
prefix = NULL,
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
currency = "EUR",
trim = TRUE,
...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks)

	• An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang <code>lambda</code> function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang <code>lambda</code> function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see <code>coord_cartesian()</code>).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function <code>expansion()</code> to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang <code>lambda</code> function notation. • The default (<code>scales::censor()</code>) replaces out of bounds values with NA. • <code>scales::squish()</code> for squishing out of bounds values into range. • <code>scales::squish_infinite()</code> for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	<code>sec_axis()</code> is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .

trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>scale_(x y)_continuous()</code>
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_pt()
scale_x_percent_pt()
scale_x_currency_pt()
```

scale-ru

Russian Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_ru(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_percent_ru(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_x_currency_ru(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    currency = "RUB",
    trim = TRUE,
    ...
)

scale_y_number_ru(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_percent_ru(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_currency_ru(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "RUB",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for po-

	sition scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • NULL for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code>. Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range.

	• <code>scales::squish_infinite()</code> for squishing infinite values into range.
<code>na.value</code>	Missing values will be replaced with this value.
<code>trans</code>	[Deprecated] Deprecated in favour of <code>transform</code> .
<code>position</code>	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
<code>sec.axis</code>	<code>sec_axis()</code> is used to specify a secondary axis.
<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
<code>scale</code>	A scaling factor: <code>x</code> will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
<code>prefix</code>	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
<code>suffix</code>	Additional text to display after the number.
<code>big.mark</code>	Character used between every 3 digits to separate thousands. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>decimal.mark</code>	The character to be used to indicate the numeric decimal point. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>trim</code>	Logical, if <code>FALSE</code> , values are right-justified to a common width (see <code>base::format()</code>).
<code>...</code>	Other arguments passed on to <code>scale_(x y)_continuous()</code>
<code>currency</code>	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A `ggplot2` scale, for use in a `+`-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_ru()
scale_x_percent_ru()
scale_x_currency_ru()
```

scale-sa

Saudi Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_sa(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_sa(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_currency_sa(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "SAR",
  trim = TRUE,
  ...
)

scale_y_number_sa(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```
    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_percent_sa(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_sa(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

position = "left",
sec.axis = waiver(),
accuracy = 1,
scale = 1,
prefix = NULL,
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
currency = "SAR",
trim = TRUE,
...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks)

	• An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang <code>lambda</code> function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang <code>lambda</code> function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see <code>coord_cartesian()</code>).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function <code>expansion()</code> to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang <code>lambda</code> function notation. • The default (<code>scales:::censor()</code>) replaces out of bounds values with NA. • <code>scales:::squish()</code> for squishing out of bounds values into range. • <code>scales:::squish_infinite()</code> for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	<code>sec_axis()</code> is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .

trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>scale_(x y)_continuous()</code>
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_sa()
scale_x_percent_sa()
scale_x_currency_sa()
```

scale-se

Swedish Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_se(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_percent_se(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = "",
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_x_currency_se(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    currency = "SEK",
    trim = TRUE,
    ...
)

scale_y_number_se(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_percent_se(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = "",
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_currency_se(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "SEK",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for po-

	sition scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • NULL for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code>. Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range.

	• <code>scales::squish_infinite()</code> for squishing infinite values into range.
<code>na.value</code>	Missing values will be replaced with this value.
<code>trans</code>	[Deprecated] Deprecated in favour of <code>transform</code> .
<code>position</code>	For position scales, The position of the axis. <code>left</code> or <code>right</code> for y axes, <code>top</code> or <code>bottom</code> for x axes.
<code>sec.axis</code>	<code>sec_axis()</code> is used to specify a secondary axis.
<code>accuracy</code>	A number to round to. Use (e.g.) <code>0.01</code> to show 2 decimal places of precision. If <code>NULL</code> , the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
<code>scale</code>	A scaling factor: <code>x</code> will be multiplied by <code>scale</code> before formatting. This is useful if the underlying data is very small or very large.
<code>prefix</code>	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) <code>-\$1</code> and <code>(\$1)</code> .
<code>suffix</code>	Additional text to display after the number.
<code>big.mark</code>	Character used between every 3 digits to separate thousands. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>decimal.mark</code>	The character to be used to indicate the numeric decimal point. The default (<code>NULL</code>) retrieves the setting from the number options .
<code>trim</code>	Logical, if <code>FALSE</code> , values are right-justified to a common width (see <code>base::format()</code>).
<code>...</code>	Other arguments passed on to <code>scale_(x y)_continuous()</code>
<code>currency</code>	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A `ggplot2` scale, for use in a `+`-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_se()
scale_x_percent_se()
scale_x_currency_se()
```

scale-tr	<i>Turkish Style Positional Scales</i>
----------	--

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_tr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",  
  sec.axis = waiver(),  
  accuracy = 1,  
  scale = 1,  
  prefix = "",  
  suffix = "",  
  big.mark = NULL,  
  decimal.mark = NULL,  
  trim = TRUE,  
  ...  
)
```

```
scale_x_percent_tr(  
  name = waiver(),  
  breaks = waiver(),  
  minor_breaks = waiver(),  
  guide = waiver(),  
  n.breaks = NULL,  
  labels = waiver(),  
  limits = NULL,  
  expand = waiver(),  
  oob = censor,  
  na.value = NA_real_,  
  trans = "identity",  
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_currency_tr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "TRY",
  trim = TRUE,
  ...
)

scale_y_number_tr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```
    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_percent_tr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_tr(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

position = "left",
sec.axis = waiver(),
accuracy = 1,
scale = 1,
prefix = NULL,
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
currency = "TRY",
trim = TRUE,
...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks)

	• An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang <code>lambda</code> function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang <code>lambda</code> function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see <code>coord_cartesian()</code>).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function <code>expansion()</code> to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang <code>lambda</code> function notation. • The default (<code>scales::censor()</code>) replaces out of bounds values with NA. • <code>scales::squish()</code> for squishing out of bounds values into range. • <code>scales::squish_infinite()</code> for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	<code>sec_axis()</code> is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .

trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>scale_(x y)_continuous()</code>
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_tr()
scale_x_percent_tr()
scale_x_currency_tr()
```

scale-us	<i>US Style Positional Scales</i>
----------	-----------------------------------

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_us(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_number_us(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_x_percent_us(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_percent_us(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_x_currency_us(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```

    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    currency = "USD",
    trim = TRUE,
    ...
)

scale_y_currency_us(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "USD",
  trim = TRUE,
  ...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for po-

	sition scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • NULL for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code>. Use NULL to use the default number of breaks given by the transformation.
labels	One of: • NULL for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks) • An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang lambda function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang lambda function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the <code>limit</code> argument in the coordinate system (see coord_cartesian()).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function expansion() to generate the values for the <code>expand</code> argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang lambda function notation. • The default (scales::censor()) replaces out of bounds values with NA. • scales::squish() for squishing out of bounds values into range.

	• scales::squish_infinite() for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	sec_axis() is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before style_positive and style_negative are processed so that prefix = "\$" will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .
trim	Logical, if FALSE, values are right-justified to a common width (see base::format()).
...	Other arguments passed on to scale_(x y)_continuous()
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
demo_number(c(-1e6, 1e6), scale_name = "number_us")
scale_x_number_us()
demo_number(c(-1, 1), scale_name = "percent_us")
scale_x_percent_us()
demo_number(c(-1e4, 1e4), scale_name = "currency_us")
scale_x_currency_us()
```

scale-za

South African Style Positional Scales

Description

The `scale_xxx_xxx` family of functions makes it easy to format axis text in decimal format, as percentages or as currencies.

Usage

```
scale_x_number_za(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = "",
  suffix = "",
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)
```

```
scale_x_percent_za(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
```

```
    sec.axis = waiver(),
    accuracy = 1,
    scale = 100,
    prefix = NULL,
    suffix = NULL,
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_x_currency_za(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "bottom",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 1,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  currency = "ZAR",
  trim = TRUE,
  ...
)

scale_y_number_za(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```
    position = "left",
    sec.axis = waiver(),
    accuracy = 1,
    scale = 1,
    prefix = "",
    suffix = "",
    big.mark = NULL,
    decimal.mark = NULL,
    trim = TRUE,
    ...
)

scale_y_percent_za(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
  position = "left",
  sec.axis = waiver(),
  accuracy = 1,
  scale = 100,
  prefix = NULL,
  suffix = NULL,
  big.mark = NULL,
  decimal.mark = NULL,
  trim = TRUE,
  ...
)

scale_y_currency_za(
  name = waiver(),
  breaks = waiver(),
  minor_breaks = waiver(),
  guide = waiver(),
  n.breaks = NULL,
  labels = waiver(),
  limits = NULL,
  expand = waiver(),
  oob = censor,
  na.value = NA_real_,
  trans = "identity",
```

```

position = "left",
sec.axis = waiver(),
accuracy = 1,
scale = 1,
prefix = NULL,
suffix = NULL,
big.mark = NULL,
decimal.mark = NULL,
currency = "ZAR",
trim = TRUE,
...
)

```

Arguments

name	The name of the scale. Used as the axis or legend title. If <code>waiver()</code> , the default, the name of the scale is taken from the first mapping used for that aesthetic. If <code>NULL</code> , the legend title will be omitted.
breaks	One of: • <code>NULL</code> for no breaks • <code>waiver()</code> for the default breaks computed by the transformation object • A numeric vector of positions • A function that takes the limits as input and returns breaks as output (e.g., a function returned by <code>scales::extended_breaks()</code>). Note that for position scales, limits are provided after scale expansion. Also accepts rlang lambda function notation.
minor_breaks	One of: • <code>NULL</code> for no minor breaks • <code>waiver()</code> for the default breaks (one minor break between each major break) • A numeric vector of positions • A function that given the limits returns a vector of minor breaks. Also accepts rlang lambda function notation. When the function has two arguments, it will be given the limits and major breaks.
guide	A function used to create a guide or its name. See guides() for more information.
n.breaks	An integer guiding the number of major breaks. The algorithm may choose a slightly different number to ensure nice break labels. Will only have an effect if <code>breaks = waiver()</code> . Use <code>NULL</code> to use the default number of breaks given by the transformation.
labels	One of: • <code>NULL</code> for no labels • <code>waiver()</code> for the default labels computed by the transformation object • A character vector giving labels (must be same length as breaks)

	• An expression vector (must be the same length as breaks). See <code>?plotmath</code> for details. • A function that takes the breaks as input and returns labels as output. Also accepts rlang <code>lambda</code> function notation.
limits	One of: • NULL to use the default scale range • A numeric vector of length two providing limits of the scale. Use NA to refer to the existing minimum or maximum • A function that accepts the existing (automatic) limits and returns new limits. Also accepts rlang <code>lambda</code> function notation. Note that setting limits on positional scales will remove data outside of the limits. If the purpose is to zoom, use the limit argument in the coordinate system (see <code>coord_cartesian()</code>).
expand	For position scales, a vector of range expansion constants used to add some padding around the data to ensure that they are placed some distance away from the axes. Use the convenience function <code>expansion()</code> to generate the values for the expand argument. The defaults are to expand the scale by 5% on each side for continuous variables, and by 0.6 units on each side for discrete variables.
oob	One of: • Function that handles limits outside of the scale limits (out of bounds). Also accepts rlang <code>lambda</code> function notation. • The default (<code>scales::censor()</code>) replaces out of bounds values with NA. • <code>scales::squish()</code> for squishing out of bounds values into range. • <code>scales::squish_infinite()</code> for squishing infinite values into range.
na.value	Missing values will be replaced with this value.
trans	[Deprecated] Deprecated in favour of transform.
position	For position scales, The position of the axis. left or right for y axes, top or bottom for x axes.
sec.axis	<code>sec_axis()</code> is used to specify a secondary axis.
accuracy	A number to round to. Use (e.g.) 0.01 to show 2 decimal places of precision. If NULL, the default, uses a heuristic that should ensure breaks have the minimum number of digits needed to show the difference between adjacent values. Applied to rescaled data.
scale	A scaling factor: x will be multiplied by scale before formatting. This is useful if the underlying data is very small or very large.
prefix	Additional text to display before the number. The suffix is applied to absolute value before <code>style_positive</code> and <code>style_negative</code> are processed so that <code>prefix = "\$"</code> will yield (e.g.) -\$1 and (\$1).
suffix	Additional text to display after the number.
big.mark	Character used between every 3 digits to separate thousands. The default (NULL) retrieves the setting from the number options .
decimal.mark	The character to be used to indicate the numeric decimal point. The default (NULL) retrieves the setting from the number options .

trim	Logical, if FALSE, values are right-justified to a common width (see <code>base::format()</code>).
...	Other arguments passed on to <code>scale_(x y)_continuous()</code>
currency	currency symbol

Details

- the number ones format axis text in decimal format.
- the percent ones format axis text as percentages.
- the currency ones format axis text as currencies.

Value

A ggplot2 scale, for use in a +-chained plot construction.

Examples

```
# not rendered via demo_number(): several locales' group/currency signs
# aren't in the check device's font, so only the (unrendered) scale
# objects are shown
scale_x_number_za()
scale_x_percent_za()
scale_x_currency_za()
```

show_locales	<i>Locales</i>
--------------	----------------

Description

A convenience function which returns a dataframe of available locales and countries.

Usage

```
show_locales()
```

Value

a dataframe

Examples

```
show_locales()
```

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